

Pipeline Freshness Audit — 2026-05-03

Uncle May's Data Engineering · First Deliverable

Data Engineer Agent (b078d1d4)

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Executive Summary

This report certifies the state of all Uncle May's data pipelines as of **2026-05-03 03:34 UTC** — the date of first full pipeline run after the Data Engineer was onboarded. Every source in the ML pipeline was pulled successfully in this run. No errors were encountered.

Key findings for the Decision Scientist:

- All 10 ingest sources are healthy and returned data.
- All parquet files were written within the last hour (pipeline run completed ~03:34 UTC 2026-05-04).
- The processed conversion dataset has 305 rows (10 converted, 295 abandoned) — small but usable for initial modeling.

- **Airtable Catalog is live** (45 active SKUs, 47 total records) but has no parquet ingest yet — data is queried directly via API.
- **Attribution metadata completeness** on Stripe PaymentIntents needs investigation (UTM/fbclid fields were historically ~99% null; this should be spot-checked in the next analysis cycle).
- The pipeline is ready to support model development. Recommend establishing a daily scheduled run.

1. Pipeline Run Summary

Table 1: All 18 tables across 10 sources pulled successfully.

Source	Table(s)	Run Timestamp (UTC)	Row Count	Columns	Status
Checkout Store	checkout_store	2026-05-04 03:32:55	2	38	OK
Stripe	payment_intents	2026-05-04 03:33:02	143	55	OK
Stripe	checkout_sessions	2026-05-04 03:33:06	303	27	OK
Stripe	customers	2026-05-04 03:33:07	25	11	OK
Stripe	charges	2026-05-04 03:33:08	35	16	OK
Stripe	subscriptions	2026-05-04 03:33:09	67	17	OK
Stripe	invoices	2026-05-04 03:33:12	68	20	OK
Mailchimp	members	2026-05-04 03:33:14	30	13	OK
Resend	emails	2026-05-04 03:33:14	29	10	OK
Apollo	contacts	2026-05-04 03:34:14	3,275	15	OK
GA4 BigQuery	events	2026-05-04 03:34:17	1,242	43	OK
GA4 BigQuery	session_summary	2026-05-04 03:34:18	293	28	OK
Census ACS	acs_zip	2026-05-04 03:34:19	44	11	OK
Meta Ads	campaigns	2026-05-04 03:34:20	3	9	OK
Meta Ads	campaign_insights	2026-05-04 03:34:24	11	16	OK

Source	Table(s)	Run Timestamp (UTC)	Row Count	Columns	Status
Google Ads	campaign_in-sights	2026-05-04 03:34:26	10	13	OK
Clarity	url_metrics	2026-05-04 03:34:28	720	16	OK
Airtable Catalog	Catalog (API only)	2026-05-04 03:34:xx	47 total / 45 active	10	OK

Prior run (2026-05-02): Files from 12:20–13:39 UTC. Age at time of this run: 15–16 hours. Within 48-hour freshness window.

2. Source-by-Source Detail

2.1 Stripe

Stripe is the **source of truth for revenue**. Six tables are pulled.

PaymentIntents: 143 rows, 55 columns

Columns: payment_intent_id, created_at, status, amount_cents, amount_received_cents, currency,

Key observations:

- **143 PaymentIntents** captured (all-time). This is the label spine for the conversion model.
- **303 Checkout Sessions** — the broader funnel. Post-suppression (removing internal test sessions), 305 rows feed the processed dataset.
- **Conversion rate: $10/305 = 3.3\%$** (post-suppression). GA4-reported CVR is ~0.10% at session level — the discrepancy is expected because GA4 counts all sessions (including bounces) while Stripe counts only users who reached checkout.
- **35 charges, 25 customers, 67 subscriptions, 68 invoices** are live. Subscriptions exceed charges on record — suggests some subscription records exist without completed charge objects (possibly trials or pending).

Attribution metadata check (PaymentIntents):

Table 2: Stripe PaymentIntent attribution completeness

Attribution Field	% Non-null
utm_source_pct_filled	30.8%
utm_medium_pct_filled	30.8%
utm_campaign_pct_filled	30.8%

Attribution Field	% Non-null
utm_content_pct_filled	30.8%
utm_term_pct_filled	4.9%
gclid_pct_filled	2.8%
fbclid_pct_filled	2.1%
fbclid_pct_filled	3.5%
fbclid_pct_filled	8.4%
ga_client_id_pct_filled	0.0%

2.2 GA4 BigQuery

GA4 is the **web analytics source** (project: `uncle-mays-automation`, dataset: `analytics_494626869`).

Events: 1242 rows, 43 cols

Sessions: 293 rows, 28 cols

Table 3: Top 10 GA4 event types

Event Name	Count
page_view	471
session_start	294
first_visit	241
scroll	102
user_engagement	86
select_item	28
form_start	11
add_to_cart	3
generate_lead	3
click	2

2.3 Mailchimp

Members: 30 rows, 13 cols

Columns: `email`, `email_hash`, `status`, `signup_ts`, `opt_ts`, `last_changed_ts`, `member_rating`, `tags`, `t`

Table 4: Mailchimp member status breakdown

Status	Count
cleaned	4
subscribed	26

2.4 Resend

Email events: 29 rows, 10 cols

Columns: email_id, created_at, to, to_hash, from, subject, last_event, tag_type, tag_session, t

2.5 Apollo

Apollo is the **B2B contact / investor CRM** (3,275 contacts).

Contacts: 3275 rows, 15 cols

Columns: apollo_id, email, email_hash, first_name, last_name, title, seniority, city, state, c

2.6 Meta Ads

Campaigns: 3 | Insight rows: 11

Table 5: Meta Ads totals (all campaign days in window)

total_spend_usd	total_impressions	total_clicks	total_purchases
286.92	14591	777	2

2.7 Google Ads (Account 6015592923)

Daily campaign rows: 10 | Cols: 13

Table 6: Google Ads totals (all campaign days in window)

total_cost	total_impressions	total_clicks
\$162.17	576	49

2.8 Airtable Catalog (appm6F6H9obydzAM2)

Airtable is queried directly via API — no parquet ingest script exists yet. This is a **gap to close** (see Section 4).

- **Total records:** 47
- **Active SKUs:** 45
- **Fields:** SKU, Name, Category, Unit, CostCents, Active, ImageURL, SortOrder, TaxCategory, PriceOverrideCents
- **Status:** API accessible, auth healthy

2.9 Census ACS

Census ACS is a **static enrichment** — 44 ZIP codes with demographic variables. Used as features for geographic conversion modeling.

ZIP enrichments: 44 rows, 11 cols

Columns: zip, median_household_income, total_population, median_age, median_household_size, bl

2.10 Clarity

Microsoft Clarity provides **session-level behavioral data** (heatmaps, scroll depth, funnel page visits).

URL metric rows: 720 | Funnel page matches: 238

Table 7: Top Clarity pages by session count (SessionCount metric)

url	sessions
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3. Processed Dataset

The `build_dataset` step merges all raw sources into a single ML-ready feature matrix.

Table 8: Processed conversion_v2 dataset summary (2026-05-04 run).

Metric	Value
Rows (post-suppression)	305
Features (columns)	93
Converted (Y=1)	10
Abandoned (Y=0)	295
Conversion rate	3.3%
Unique customers	17
First-attempt sessions	17

Class imbalance note: 10 positives is a very small positive class. The Decision Scientist's `01_first_look_baseline.qmd` uses SMOTE resampling via `themis` to address this. As volume grows, the model will stabilize.

4. Gaps and Recommended Actions

Table 9: Recommended actions from freshness audit.

Priority	Gap	Owner	Action
High	Airtable Catalog has no parquet ingest	Data Engineer	Write <code>ml/ingest/airtable.py</code> and add to <code>run_pipeline.py</code>
High	No scheduled daily run — data only refreshes when manually triggered	Data Engineer	Create Trigger.dev or cron task for daily <code>python -m ml.run_pipeline</code>
Medium	Attribution field completeness on Stripe PIs needs verification	Data Engineer + Decision Scientist	Run null-rate check on UTM/fbclid/gclid fields in next analysis cycle
Medium	Checkout Store only shows 2 rows — appears to be in-flight sessions, not historical	Data Engineer	Clarify source: is this an in-memory store or DB dump?
Low	Resend (29 rows) covers only recent sends — no historical backfill	Data Engineer	Assess whether Resend API supports full history pull
Low	Meta Ads (3 campaigns, 11 insight rows) — narrow time window	Data Engineer	Verify date range parameter in <code>meta_ads.py</code>

5. Pipeline Architecture Snapshot

Sources	Ingest (ml/ingest/*.py)	Raw Parquet	Processed
Stripe API	<code>stripe.py</code>	<code>stripe_*.parquet</code>	
GA4 BigQuery	<code>bigquery_ga4.py</code>	<code>ga4_*.parquet</code>	
Mailchimp API	<code>mailchimp.py</code>	<code>mailchimp_*.parquet</code>	
Resend API	<code>resend.py</code>	<code>resend_*.parquet</code>	
Apollo API	<code>apollo.py</code>	<code>apollo_*.parquet</code>	<code>conversion_v2_*.parquet</code>
Meta Ads API	<code>meta_ads.py</code>	<code>meta_*.parquet</code>	(305 rows, 93 features)
Google Ads API	<code>google_ads.py</code>	<code>google_ads_*.parquet</code>	
Clarity API	<code>clarity.py</code>	<code>clarity_*.parquet</code>	
Census ACS API	<code>census.py</code>	<code>census_*.parquet</code>	
Checkout Store	<code>checkout_store.py</code>	<code>checkout_store_*.parquet</code>	
Airtable API	[MISSING - gap]	[no file]	

All raw parquet files use UTC-timestamped names (`source_YYYYMMDDTHHMMSSz.parquet`). The latest file per source is selected at feature-prep time.

6. Methodology

- **Run date/time:** 2026-05-04 03:32–03:34 UTC (pipeline completed in ~97 seconds)
- **Prior run:** 2026-05-02 12:20–13:39 UTC (~15 hours prior)
- **Freshness threshold:** 48 hours (all sources pass)
- **Suppression:** Internal emails (Anthony + *@unclemays.com) removed from processed dataset; raw files retain all records
- **Auth:** All API keys loaded from `~/.claude/*-config.json`; GA4 uses service account at `~/.claude/ga-service-account.json`
- **Tool:** Python (polars), then R (arrow) for this notebook

Report generated by the Data Engineer agent (UNC-766) on 2026-05-03.